

Victor Mororó

Software Developer

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SUMMARY

An undergraduate student in Computer Engineering with a strong focus on combinatorial optimization, performance optimization, low-level programming, and graphics programming. Proficient in multiple programming languages, with a solid drive to solve complex problems through efficient algorithms and software development, and a deep interest in exploring the fundamental concepts behind these areas.

EDUCATION

2020 – Present **BSc in Computer Engineering** at [UFPB](#) (8.73/10)

WORK EXPERIENCE

Undergraduate Researcher at [LOG – UFPB](#) Jul 2021 - Present

- Developing a study aimed at conducting a comparative analysis of multiple programming languages in solving the Minimum Latency Problem using the GILS-RVND heuristic. Languages include C, C++, Rust, Fortran, Julia, Python, Java, JavaScript, Lua, MATLAB, and Go. [Source](#)

Undergraduate Teaching Assistant | Introduction to Computer Engineering Aug 2021 - Dec 2021

- Provided ongoing support in practical activities, theoretical understanding, and exam preparation.
- Facilitated the transition to remote learning.
- Recognized for contributions at the university's teaching initiation event (XXIII ENID).

PROJECTS

draw | CPU-based Real-time renderer (*in development*) [Source](#)

- Developed in Rust, *from scratch*, this project implements triangle rasterization, perspective projection, Phong shading, and texture mapping, operating without a graphics API by drawing directly in the frame buffer using the X11 API.

Optimization Kit Implementations | LOG – UFPB [Source](#)

- Personal implementations of various combinatorial optimization methods for solving the Traveling Salesman Problem (TSP) and the Bin Packing Problem (BPP), including Branch-and-Bound, Lagrangian Relaxation, Branch-and-Cut, and Branch-and-Price, as well as the GILS-RVND heuristic.

DSanim | Data Structures visualizer [Source](#)

- Interactive tool to visualize data structures through animations using Python.
- Utilizes the Manim library for rendering animations, GStreamer for handling multimedia, and GTK for the graphical user interface.

AWARDS

2021 **Undergraduate Teaching Award** at XXIII ENID – UFPB [Article](#)

SKILLS AND INTERESTS

Programming Languages C, C++, Rust, Python, Julia, Fortran, x86-64 Assembly, Java.

Tools Linux, Gdb, Git, Valgrind, Makefile, LaTeX, Vim.

Interests Systems Programming, Performance Optimization, Combinatorial Optimization, Graphics Programming.

Languages Portuguese (Native), English (Intermediate).